

# Eddie Tang

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## EDUCATION

### University of Illinois Urbana-Champaign

Urbana, IL

*BS in Electrical Engineering | James Scholar Honors | GPA: 4.0*

*Aug. 2025 – May 2028*

*Minor in Computer Science*

## PROJECTS & LEADERSHIP

### CAM-MK3 | Illinois Space Society - Spaceshot Avionics | KiCAD, C++, ESP32-IDF Oct. 2025 – Present

- Created compact live-video transmitter & dual-camera controller PCBA for rocket avionics; designed schematic, layout, and power architecture to meet size and RF link budget constraints.
- Authored and validated firmware drivers for NTSC→digital video, hardware JPEG encoding, & radio transceiving
- Implemented B2B I<sup>2</sup>C control and 433 MHz wireless communication for telemetry commands and live-video feed
- Built and debugged multiple revisions and validated signal & power integrity with oscilloscope and bench power supplies. Flown built boards on rockets up to 13,455ft and 30,994ft.

### Project EMBER (Technical Lead) | Illini Solar Car | KiCAD, C++, Simplicity Studio Jan. 2026 – Present

- Spearheaded Project EMBER, a joint initiative with AbbVie to investigate ultra-low-power design and implementation for space-constrained PCBs, through the development of a TPMS system.
- Facilitated design, development, & bring-up of schematic and layout of a 17mm-diameter TPMS sensor node PCB
- Developed custom FSM and BLE payload & protocol to consume ~4μA when idle and 168μA average when active, allowing continuous use of up to 135 days
- Coordinated deliverables, weekly meetings, KPIs, timeline, and project reports with AbbVie by delegating work across an 8-member team spanning firmware and receiver node layout.

### GATOR | Illinois Space Society - Spaceshot Avionics | KiCAD, C++ Oct. 2025 – Present

- Implemented & designed GATOR, a power-distribution PCB to drive 24V servos to dynamically control the orientation of a yagi antenna according to relative rocket position.
- Programmed GATOR orientation feedback loop using GPS data from LoRa telemetry packets and servo angle

### TARC Avionics V1–V3 (Lead & Founder) | EasyEDA, C++, Python Jun. 2022 – Jun. 2025

- Designed, assembled, and programmed custom 6-DOF AHRS PCB using BMI088 IMU and MS5607 barometer for active flight control to variable target apogees with 80% success rate
- Developed quaternion-based sensor fusion algorithm for accurate altitude & velocity estimation from acceleration
- Built Python software-in-the-loop simulations to validate control logic using historical flight telemetry
- Ranked top 50 at TARC Nationals (from pool of 1,000 teams); qualified twice for Nationals

## EXPERIENCE

### Launch Operations Member | Illinois Space Society - Spaceshot Avionics Sept. 2025 - Present

- Manned, assembled, and configured flight ground stations for 3 high-power rocket launches (>20k ft), including LoRa/RF link verification, telemetry receiver initialization, and antenna pointing before each launch window.
- Conducted pad procedures & AV-bay integration, including continuity checks on pyros and arming sequences
- Monitored real-time telemetry data during flight and recovery phases to support post-flight vehicle recovery.

### Research Assistant (Paid), Mechanical Engineering | Rowan University Summer 2025

- Investigated effects of point defects in thermoelectric materials on lattice thermal conductivity using MD.
- First author on in-review manuscript investigating point defect effects on thermal conductivity in Si and Ge.

### Laboratory Learning Program, Mechanical & Aerospace Department | Princeton University Summer 2024

- Performed DFT calculations on WO<sub>3</sub> surfaces for plasma-assisted NH<sub>3</sub> synthesis; analyzed results with NumPy.
- Co-authored a peer-reviewed [publication](#) in ACS Energy Letters (Impact Factor: 19.3).

## SKILLS

**Design Tools:** KiCAD, EAGLE, Vivado, OnShape, NX Siemens, Git

**Languages:** C++, Python, C, Bash, Dart, Go, Java, MATLAB

**Protocols:** I2C, SPI, UART, CAN, Bluetooth, WiFi, LoRa

**Lab Equipment:** Oscilloscope, Function Generator, Spectrum Analyzer, Software Defined Radio, RF Amplifier

**Other:** HAM Radio Technician (KE2CNQ); Fluent in English and Chinese; NMSQT Finalist; 1560 SAT